

Rare Coagulation Factor Deficiencies (Factors VII, X, V, and II)



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KEYWORDS

- Rare bleeding disorders • Epistaxis • Factor II • Factor V • Factor VII • Factor X
- Heavy menstrual bleeding

KEY POINTS

- Rare coagulation factor deficiencies, more commonly referred to as rare bleeding disorders (RBD), comprise a rare group of disorders that can result in a variety of bleeding symptoms.
- Abnormalities in clotting times and low plasma factor activity levels help guide the diagnosis of factor II, V, VII, and X deficiencies.
- The most common bleeding manifestations in RBD are epistaxis and heavy menstrual bleeding, although intracranial hemorrhage and umbilical cord bleeding can occur in neonates with severe factor deficiencies.
- The ideal treatment modality for RBD are high-purity, single-factor products for each disorder, but alternative treatment modalities exist for factor II and V deficiencies that lack such a product.

INTRODUCTION

von Willebrand disease, hemophilia A, and hemophilia B account for most inherited coagulation factor deficiencies worldwide. Inherited rare coagulation factor deficiencies, more commonly referred to as rare bleeding disorders (RBD), consist of deficiencies in fibrinogen as well as factors II (FII), V (FV), VII (FVII), X (FX), XI (FXI), XIII (FXIII); combined factor V + VIII deficiency; and vitamin K-dependent coagulation factor deficiency. Inherited RBD represent only 3% to 5% of all inherited deficiencies of coagulation factors. However, individuals with RBD can present with clinically significant bleeding and are at risk of delayed diagnosis or misdiagnosis due to the rare

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nature of these disorders.¹ RBD are predominantly inherited in an autosomal recessive pattern and are more common in areas with higher rates of consanguineous marriages. The heterozygous state is usually asymptomatic. This review focuses on deficiencies of coagulation FII, FV, FVII, and FX.

Pathophysiology

Factor II (prothrombin) deficiency: The American physician and biochemist Dr Armand J. Quick^{2,3} first described FII deficiency in 1947 after observing prolonged prothrombin times (PT) and low FII activity in 2 unrelated families. Prothrombin is a 72-kDa glycoprotein synthesized in the liver in the presence of vitamin K and circulates as an inactive zymogen in the plasma. Prothrombin is activated into the active enzyme thrombin when cleaved by the activated factor X (FXa)-activated factor V (FVa) prothrombinase complex on the phospholipid-rich surface of platelets. Thrombin is a critical factor in multiple pathways of clot formation through activation of platelet aggregation and clot potentiation by activation of FV, FVIII, and FIX that forms a positive feedback loop resulting in further thrombin generation.⁴ Furthermore, thrombin helps to stabilize fibrin clot formation through the activation of FXIII and thrombin activatable fibrinolysis inhibitor (TAFI). Thrombin also plays a role in mechanisms of anticoagulation. Thrombin interacts with endothelial protein thrombomodulin to activate protein C, which in turn inactivates FVa and activated FVIII (FVIIIa) to reduce clot formation. Thrombin regulation is important for preventing excessive clot formation. Serine protease inhibitor (serpin) family members, such as heparin cofactor II and protease nexin I, inhibit the catalytic activity of thrombin.⁵ Antithrombin, a glycoprotein produced in the liver, can inhibit thrombin and FXa activity when coupled with heparin.⁶

FV deficiency: Dr Paul Owren first described FV deficiency in a Norwegian woman with severe recurrent epistaxis and heavy menstrual bleeding in 1943.⁷ FV is a 330-kDa glycoprotein produced in the liver that circulates in plasma. Approximately 20% of FV is stored in the alpha granules of platelets and can be released at sites of vascular injury with platelet degranulation.⁸ FV is activated via proteolytic cleavage by thrombin and FXa. FV plays an important role in the propagation phase of coagulation as a component of the prothrombinase complex. FVa complexes with FXa on a phospholipid surface in the presence of calcium to form the prothrombinase complex that cleaves FII into thrombin.⁹ Thrombin bound to thrombomodulin activates protein C, which binds protein S, to inactivate FVa and FVIIIa through proteolytic cleavage.¹⁰

Factor VII deficiency: FVII is a 50-kDa vitamin K-dependent serine protease produced in the liver. FVII plays a critical role in the initiation of clot formation as part of the extrinsic pathway of the coagulation cascade. Upon endothelial injury, tissue factor (TF) is exposed from the vascular lumen where it can activate FVII into activated FVII (FVIIa) forming a TF/FVIIa complex. The TF/FVIIa complex is able to activate FX into FXa (forming the "extrinsic tenase" complex) and FIX into activated FIX (FIXa).¹¹ This process ultimately results in thrombin generation and fibrin clot formation. The thrombin generated during this process activates FV and FVIII through a positive feedback loop that propagates fibrin clot formation. A small portion of FVII circulates as the active serine protease FVIIa in the absence of active clot formation, which is unique to FVII; this is important because it is circulating plasma FVIIa that first engages TF at the site of injury. Furthermore, FVII can be proteolytically cleaved and activated by the TF/FVIIa complex, thrombin, FIXa, and FXa. FVIIa is primarily inhibited by the lipoprotein tissue factor pathway inhibitor (TFPI).

FX deficiency: FX is a 59-kDa vitamin K-dependent serine protease synthesized in the liver that plays a key role in coagulation as the first enzyme in the common pathway

in the formation of a stable fibrin clot. In 1905, Morawitz identified a factor named *thromboplastin* that interacted with *thrombogen* to form thrombin.¹² In 1955, Duckert and colleagues¹³ reported a factor deficiency that was distinct from FVII and FIX deficiencies in patients receiving coumarins. Inherited FX deficiency was later identified by 2 independent groups, each of which described a patient (Prower in the United Kingdom and Stuart in the United States) with a bleeding diathesis that could not be attributed to other known coagulation factors deficiencies. The factor in both patients was subsequently named FX. As with other vitamin K-dependent proteins, FX requires posttranslational carboxylation of 11 glutamic acid (Glu) residues for functional activity.^{14,15} Both the extrinsic and intrinsic pathways can lead to FX activation. In the extrinsic pathway, FX activation occurs via formation of the TF/FVIIa complex with calcium ions on a phospholipid surface. In the intrinsic pathway, FX activation occurs when FIXa and its cofactor FVIIIa form the intrinsic tenase complex in the presence of calcium ions on a phospholipid surface. FXa is the most important activator of prothrombin, cleaving prothrombin to generate thrombin through the prothrombinase complex. FXa also activates FV and FVIII and hydrolyzes FVII to FVIIa, completing the FVII-FX feedback loop. FXa also forms a quaternary complex with TF/FVIIa by binding to TFPI, which downregulates the extrinsic pathway to thrombin generation. Antithrombin inactivates FXa by forming a complex that is rapidly cleared from the circulation. The protein Z-dependent protease inhibitor, a serpin, also inactivates FXa. Defects in protein Z lead to increased FXa activity and may increase the risk of thrombosis.¹⁶

Genetics and Disease Classification

FII deficiency: The *F2* gene is encoded on chromosome 11p11-q12 and consists of 10 exons and 8 introns. Missense mutations represent most of the more than 50 different *F2* gene variants that have been identified in patients with FII deficiency, but splicing, nonsense, and deletion/insertion mutations have been reported.¹⁷ Most individuals with FII deficiency are homozygous for the same *F2* mutation inherited in an autosomal recessive pattern as opposed to compound heterozygotes for 2 separate variants. There are 2 forms of FII deficiency: (1) a quantitative deficiency (type 1 deficiency or hypoprothrombinemia) that is characterized by equal reductions in FII activity and antigen levels and (2) a qualitative deficiency (type 2 deficiency or dysprothrombinemia) characterized by discordance in the FII activity and antigen levels in which individuals have low FII activity but normal/near-normal FII antigen levels. There are no reported cases of living individuals with a complete absence of FII.¹⁸ Moreover, an FII knockout model in mice demonstrated embryonic and neonatal lethality, suggesting that complete absence of FII is incompatible with life.^{19,20}

FV deficiency: The *F5* gene is located on chromosome 1q24.2 consisting of 25 exons. Missense, nonsense, splicing, and insertion/deletion mutations have been reported among 26 *F5* variants identified in FV-deficient patients. Half of the identified mutations are within exon 13, which encodes the B domain of FV, whereas the remaining variants are scattered throughout the A and C domains of the *F5* gene.²¹ Most cases of FV deficiency are characterized by a type 1 quantitative deficiency, but 25% of individuals with FV deficiency will have low FV activities and normal FV antigen levels consistent with a type 2 qualitative defect.^{17,21} Mouse models of FV deficiency demonstrate abnormal embryonic development and early death due to hemorrhage; however, mice with FV activities less than 1% and presence of cross-reacting material are able to survive the neonatal period.

FVII deficiency: The *F7* gene is located on chromosome 13q34 and consists of 9 exons. There have been more than 200 *F7* gene variants identified that result in FVII

deficiency in an autosomal recessive pattern. Most *F7* mutations are a result of missense mutations (~60%), but nonsense, splicing, and deletion/insertion mutations have also been reported.¹⁸ In contrast to mouse knockout models of FII and FV deficiency, the FVII-deficient knockout mouse model does not result in abnormal embryonic development; however, survival to adulthood is significantly impaired due to death from hemorrhages in the neonatal period.²²

FX deficiency: The *F10* gene is also located on chromosome 13q34 just downstream from the gene encoding for FVII. Combined FVII and FX deficiency has been reported due to large deletion of the long arm of chromosome 13 and/or partial duplication.²³ The coding sequence for FX is homologous to that of the other vitamin K-dependent proteins and is divided into 8 exons, each of which encodes a specific domain within the protein. Approximately 150 mutations have been identified to date. Most mutations (~75%) like in the other RBD are missense mutations with the remaining genetic abnormalities being deletions, nonsense mutations, and splice site mutations.²⁴ Although mutations are typically homozygous, combined heterozygous mutations resulting in moderate FX deficiency has also been reported.²⁴ The FX-deficient knockout model results in partial embryonic lethality and fatal neonatal bleeding.²⁵

DISCUSSION

Epidemiology

FII deficiency is one of the rarest inherited RBD accounting for only 1.5% of RBD.²⁶ FII deficiency has an estimated worldwide prevalence of 1 in 2,000,000.¹⁷ In addition, there seems to be a higher prevalence of FII genetic variants in individuals of Latin/Hispanic origin, and approximately 70% patients with FII deficiency are from areas such as Barcelona, Padua, Segovia, and Puerto Rico.^{27,28} Likewise, FV deficiency affects an estimated 1 in 1,000,000 individuals and represents 9% of RBD diagnoses. FVII deficiency is the most common of the RBD affecting 1 in 500,000 individuals and accounting for an estimated 38% of individuals with RBD worldwide.²⁶ Similar to FV deficiency, FX deficiency has an estimated prevalence of 1 in 500,000 to 1,000,000.²⁹ Higher rates of RBD have been reported in regions with increased rates of consanguinity such as the Middle East, North Africa, India, and Pakistan.²¹

Laboratory Evaluation

First-tier coagulation testing for RBD includes assessment of the PT and activated partial thromboplastin time (aPTT) (**Fig. 1**). In FII, FV, and FX deficiencies, the PT and aPTT will be prolonged. Isolated PT prolongation is observed in FVII deficiency. If available, PT and/or aPTT mixing studies can be helpful in differentiating a true factor deficiency (ie, observed correction of the mixing time) from a factor inhibitor (ie, persistent PT and/or aPTT prolongation despite mixing).³⁰ Specific factor activity levels should be performed in the setting of abnormal coagulation studies or a known family history to confirm the diagnosis. FII, FV, FVII, and FX activities are determined by a one-stage PT-based clotting assay. The FVII activity assay uses a thromboplastin reagent. Differences in the sensitivity of thromboplastin reagents can result in variability in the measured FVII activity level. Recombinant human and human-derived thromboplastins are considered the most reliable and are recommended in clinical laboratories.³¹ Nevertheless, it is important to consider that a normal PT and aPTT does not rule out a mild factor deficiency. Thus, specific factor levels should be obtained if there is a high index of suspicion for RBD. Factor antigen levels via an immunoassay

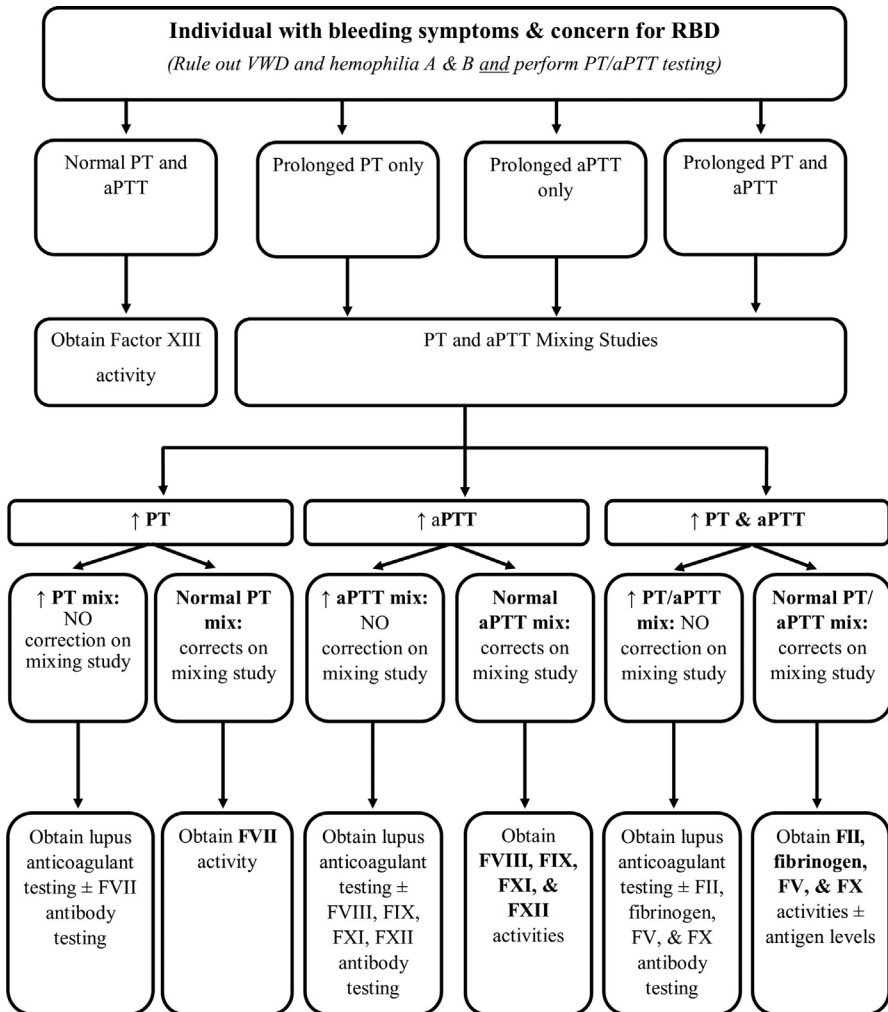


Fig. 1. Algorithm of approach to coagulation testing for RBD. Laboratory testing for an RBD should include evaluation of the complete blood cell count and complete metabolic panel to assess underlying health status and evaluate for cytopenias that could elucidate the cause of bleeding symptoms. Testing for von Willebrand disease and hemophilia A/B should be performed to rule out these more common bleeding disorders. Coagulation testing includes PT and aPTT with mixing studies (if available). Prolonged PT, aPTT, or both PT/aPTT mixing studies help guide testing for specific factor activity levels versus lupus anticoagulant and antibody testing.

can help determine whether an individual possesses a type 1 quantitative versus a type 2 qualitative factor deficiency.

FII, FV, FVII, and FX have reported plasma half-lives of 3 to 4 days, 36 hours, 4 to 6 hours, and 40 to 60 hours, respectively.¹⁷ The half-lives of coagulation factors should be taken into consideration when measuring and interpreting factor activity levels, especially following administration of hemostatic agents such as fresh frozen plasma (FFP), cryoprecipitate, or factor concentrates. A molecular diagnosis of these

deficiencies may be helpful in cases of diagnostic uncertainty or in the prenatal setting for families with a known history of RBD and a severe bleeding phenotype; even so, mutation analysis is not required for diagnosis.

Differential Diagnosis

Children with the rare acquired disorder of lupus anticoagulant hypoprothrombinemia syndrome (LAHS) can present with prolonged PT/aPTT and low FII activity similar to inherited FII deficiency. However, patients with LAHS typically present with an acute onset of bleeding symptoms after a viral illness and coagulation studies demonstrate a PT/aPTT mixing study that does not correct, detection of a lupus anticoagulant, and presence of antiprothrombin antibodies. It is important to consider LAHS in the differential diagnoses in individuals younger than 16 years with an acute onset of bleeding, history of a viral prodrome, and prolonged PT/aPTT.³² Acquired FII, FVII, and FX deficiencies can be observed in the setting of liver disease, disseminated intravascular coagulation, use of vitamin K antagonist warfarin, or vitamin K deficiency. Acquired FV deficiency secondary to an inhibitor has been described primarily in elderly individuals with comorbidities predominantly triggered by antibiotic treatment.³³ There is also report of acquired FV deficiency occurring in a patient with myeloma and amyloidosis.³⁴ Isolated acquired FVII deficiency can occur in the presence or absence of an FVII inhibitor. Acquired FVII deficiency has been reported in association with severe systemic sepsis, medications, malignancy, aplastic anemia, and posthematopoietic stem cell transplantation.^{35,36} Acquired FX deficiency occurs in up to 5% of patients with amyloidosis due to adsorption into splenic amyloid fibrils.²⁴ There have been reports of acquired FX deficiency with myeloma and other malignancies, infection, and use of sodium valproate. Acquired FX inhibitors have been identified in burns, respiratory infections, and exposure to topical thrombin.³⁷

Classification

Classification into the traditional hemophilia categories of mild, moderate, and severe can be challenging in RBD (**Table 1**). Although low factor activity levels can provide confirmation of diagnosis, factor activity levels do not always correlate with the bleeding phenotype. FII activity levels seem to demonstrate a strong correlation with bleeding severity, although data are limited due to its rarity and low prevalence of individuals with FII deficiency.³⁸ FII activity levels of 20% to 30% are considered hemostatic, and severe bleeding symptoms have been reported primarily in individuals with FII activities less than 10%.³⁹ Heterozygotes typically have FII activities between 30% and 60% compared with individuals with normal FII activities of greater than 70%. In contrast, FV deficiency has a weak association between bleeding severity and FV activity.³⁹ FV activity levels of 15% to 20% are considered hemostatic. Yet, severe bleeding symptoms of hemarthrosis and intracranial hemorrhage (ICH) have been described in FV-deficient individuals with FV activities up to 50%.⁴⁰ Heterozygotes are usually asymptomatic and have FV activities of 20% to 60%.²⁸

In FVII deficiency, individuals with FVII activities less than 10% are at increased risk of major spontaneous bleeding.³⁹ FVII activities between 15% and 20% are considered hemostatic, and heterozygotes are usually asymptomatic with FVII activities of 40% to 60%. However, the plasma FVII activity level weakly correlates with bleeding risk, even within the same individual over time. One study of European and Latin American individuals with *F7* variants showed that 19% with heterozygous FVII mutations reported bleeding symptoms.⁴¹ In FX deficiency, individuals with FX activities less than 10% are at increased risk of major spontaneous bleeding and patients with levels

Table 1**Summary of prevalence and bleeding manifestations of rare bleeding disorders factor II, factor V, factor VII, and factor X deficiencies**

Factor Deficiency	Disease Prevalence	Percentage of RBD^a	Bleeding Manifestations	Factor Activity and Disease Severity^b	Correlation of Factor Activity and Clinical Symptoms
FII	1:2,000,000	1.5%	Mucosal bleeding, soft tissue bleeding, & prolonged bleeding following trauma or surgery	Severe: <1% Moderate: 1%–10% Mild: >10% Heterozygotes: 30%–60% Hemostatic level: 20%–30%	Strong association
FV	1:1,000,000	9%	Mucosal bleeding, soft tissue bleeding, & prolonged bleeding following trauma or surgery	Severe: <1% Moderate: 1%–10% Mild: >10% Heterozygotes: 20%–60% Hemostatic level: 15%–20%	Poor association
FVII	1:500,000	38%	Epistaxis, easy bruising, gum bleeding, hemarthrosis, menorrhagia, & trauma- or surgical-induced bleeding	Severe: <10% Moderate: 10%–20% Mild: >20% Heterozygotes: 40%–60% Hemostatic level: 15%–20%	Poor association
FX	1:500,000–1,000,000	8%	Umbilical cord bleeding, hemarthroses, intracranial hemorrhage, mucosal bleeding, muscle bleeding, & hematomas	Severe: <10% Moderate: 10%–40% Mild: >40% Heterozygotes: ~50% Hemostatic level: 15%–20%	Strong association

^a The World Federation of Hemophilia (WFH) survey and EN-RBD project (1).^b *Data from* (1) Palla R, Peyvandi F, Shapiro AD. Rare bleeding disorders: diagnosis and treatment. *Blood* 2015; 125(13):2052–61 and (2) Peyvandi F, DiMichele D, Bolton-Maggs PHB, et al. Classification of rare bleeding disorders (RBDs) based on the association between coagulant factor activity and clinical bleeding severity. *J Thromb Haemost*. 2012(9):1938–43.

greater than 40% to 60% are usually asymptomatic or have mild bleeding symptoms.³¹

Clinical Manifestations

The most common bleeding manifestations in FII, FV, FVII, and FX deficiencies are mucocutaneous bleeding (ie, epistaxis and heavy menstrual bleeding) and prolonged bleeding following injury or surgery.²⁸ Neonates with RBD, particularly individuals with severe FVII and FX deficiencies, can present with umbilical stump bleeding and ICH at birth or shortly thereafter. Central nervous system morbidity in the North American Rare Bleeding Disorder (NARBD) registry was reported in 2% to 22% of these RBD.²⁸ In FX deficiency, mutations associated with ICH may justify genotyping soon after birth wherein the clinician may consider prophylaxis if present.⁴² Hemarthroses, muscle hematomas, gastrointestinal bleeding, and easy bruising have also been reported in these RBD.^{31,38} In the NARBD registry, the range of musculoskeletal complications including hemarthrosis and muscle contractures ranged from 7% in FX deficiency, 23% in FV deficiency, to 17% to 18% in FII and FVII deficiencies.²⁸ Heavy menstrual bleeding is a significant bleeding complication with a reported prevalence of 52% in women and girls with RBD.⁴³ Women with RBD are also at increased risk of bleeding during pregnancy, delivery, and postpartum and may have an increased risk of fetal loss.⁴⁴ Reproductive age issues related to RBD are further discussed later in this review.

Heterozygote carriers regardless of gender are usually asymptomatic, except for heterozygous FV, FVII, and FX carriers, of which a portion have reported bleeding symptoms.^{28,40} A study of 128 FX heterozygotes matched with an unaffected family member by gender and age noted that 29.7% had 1 or more bleeding symptom compared with 2.3% of control subjects.⁴⁵ The most common symptom was bleeding following an invasive procedure (ie, dental extraction or surgery).

Therapeutic Options

Recommendations by the European Network of Rare Bleeding Disorders (EN-RBD) project advise minimum plasma FII, FV, FVII, and FX activities of greater than 10%, 10%, greater than 20%, and 40%, respectively, as the target factor trough levels to prevent bleeding symptoms.²⁹ At present, there are no single-factor concentrates commercially available for FII- and FV-deficient individuals. There are dedicated single-factor products for FVII- and FX-deficient individuals. Recommendations for on-demand and prophylactic dosing of factor concentrates of these RBD are outlined in **Table 2**.

FFP and prothrombin complex concentrates (PCCs) are the primary treatment modalities for management of acute bleeding events, surgical hemostasis, and long-term prophylaxis in FII deficiency. PCCs contain various amounts of factors II, VII, IX, and X depending on whether it is a 3-factor PCC (contains FII, FIX, and FX) or a 4-factor PCC (contains FII, FVII, FIX, and FX). In a few countries, a virally inactivated FFP product via solvent/detergent treatment of pooled plasma is available for individuals with RBD.⁴⁶ In most countries, FFP is tested for specific viruses but not virally inactivated. When available, solvent/detergent-treated FFP, or other mechanism of viral inactivation, is preferred for treatment in this patient population.

Neither PCCs nor cryoprecipitate contain FV, thus FFP is the mainstay of treatment in FV-deficient patients to increase their FV activity levels. Platelet transfusions have been used in combination with FFP in FV deficiency because a portion of FV is stored in platelet alpha granules, but the risk of platelet alloimmunization must be considered.^{47,48}

Table 2
Recommended on-demand and prophylactic treatment of rare bleeding disorders factor II, factor V, factor VII, and factor X deficiencies

Factor Deficiency	Factor Plasma Half-Life	Recommended Factor Trough Levels ^a	On-Demand Treatment	Prophylactic Treatment ^b
FII	3–4 d	>10%	FFP ^c 15–25 mL/kg PCC (3-F or 4-F) 20–40 U/kg	FFP ^c not preferred for prophylaxis PCC (3-F or 4-F) 20–40 U/kg 1 time/wk
FV	36 h	10%	FFP ^c 15–25 mL/kg Platelets: consider transfusion as adjunct therapy to FFP in certain circumstances	FFP ^c 15–20 mL/kg 2 times/wk Platelets are not preferred for prophylaxis
FVII	4–6 h	>20%	FFP ^c not preferred pdFVII concentrate 30–40 U/kg Recombinant FVIIa 15–30 µg/kg	FFP ^c 10–20 mL/kg 2 times/wk pdFVII concentrate 30–40 U/kg 3 times/wk Recombinant FVIIa 15–30 µg/kg 3 times/wk
FX	40–60 h	>40%	FFP ^c 10–20 mL/kg PCC (3-F or 4-F) 20–40 U/kg pd-FX/FIX concentrate 10–20 U/kg pdFX concentrate 25 U/kg	FFP ^c not preferred for prophylaxis PCC (3-F or 4-F) 20–40 U/kg 2 times/wk pd-FX/FIX concentrate 10–20 U/kg 2 times/wk pdFX concentrate 25 U/kg 2 times/wk

Abbreviations: 3-F, 3-factor PCC; 4-F, 4-factor PCC; FFP, fresh frozen plasma; FVIIa, activated factor VII; PCC, prothrombin complex concentrate; pdFVII, plasma-derived FVII concentrate; pdFX, plasma-derived FX concentrate; pdFX/FIX, plasma-derived FX and FIX concentrate.

^a Data from Menegatti M, Peyvandi F. Treatment of rare factor deficiencies other than hemophilia. *Blood*. 2019;133(5):415–24 and Mumford AD, Ackroyd S, Alikhan R, et al. Guideline for the diagnosis and management of the rare coagulation disorders: a United Kingdom Haemophilia Centre Doctors' Organization guideline on behalf of the British Committee for Standards in Haematology. *Br J Haematol*. 2014;167(3):304–26.

^b Factor dose and administration frequency may vary in the pediatric population (especially in children <12 years old) to achieve the recommended factor trough levels in light of increased factor metabolism expected in this population.

^c Viral inactivated (eg, solvent/detergent-treated) FFP preferred.

Recombinant activated FVII (rFVIIa) is the preferred treatment product in FVII deficiency for severe bleeding symptoms or surgical management. However, rFVIIa infusions administered 2 to 3 times per week for prophylaxis has been used successfully despite the short half-life.⁴⁹ In some areas of the world, a plasma-derived FVII concentrate is available as an alternative treatment when rFVIIa is not available. PCCs are not recommended in FVII deficiency due to limited or no FVII.

Historically, FFP or PCCs have been used for replacement therapy in FX deficiency. However since 2015, a high-purity, plasma-derived FX (pdFX) concentrate has been available.⁵⁰ Viral inactivation of pdFX includes solvent/detergent treatment, nanofiltration, and terminal heat treatment. A phase 3 pharmacokinetic study of 16 patients with moderate to severe FX deficiency (<5%) using a pdFX dose of 25 IU/kg established a half-life of 29.4 hours and incremental recovery of 2.00 IU/dL suggesting prophylactic dosing of 1 to 2 times per week.⁵¹ In the Ten01 trial, 16 patients were infused 25 IU/kg pdFX for 187 bleeds over a minimum study period of 6 to 24 months.⁵² Subjective rating for efficacy was “excellent or good” in 98% of subjects. A small surgical study in 5 patients undergoing 7 surgeries reported 100% efficacy.⁵³ No major adverse events or onset of inhibitory antibodies have been reported. In neonates and infants, higher dosing of 70 to 80 IU/kg for on-demand treatment and prophylaxis seems necessary.⁵⁴ In 9 children with FX less than 5%, there is a recent report of its use successfully for prophylaxis and on-demand treatment with efficacy graded excellent at a pdFX regimen of ~40 IU/kg every 3 days.⁵⁵

Antifibrinolytic drugs ϵ -aminocaproic acid and tranexamic acid are commonly used and recommended for mild bleeding symptoms such as mucocutaneous bleeding and minor surgical procedures in all RBD.

Reproductive Issues

In general, women and girls with heavy menstrual bleeding have benefited from hormonal therapies including combined oral contraceptives containing estrogen, progestin-only agents, and intrauterine devices as monotherapy or adjunctive therapy with other hemostatic agents, such as antifibrinolytic drugs.^{44,56}

FII and FV levels are unchanged and remain stable throughout pregnancy and delivery. FII activity less than 20% is considered insufficient for delivery as antepartum and postpartum hemorrhage has been reported in women with FII activities in this range. Thus, administration of PCC at 20 to 40 IU/kg is advised during labor or before cesarean delivery to achieve 20% to 40% FII activity levels.³⁸ Similarly, in FV deficiency an FV activity less than 20% is considered insufficient to support a safe delivery and prevent postpartum hemorrhage. FFP administration at 15 to 25 mL/kg to achieve a minimum FV activity of 20% to 40% at time of established labor or before cesarean section is recommended.³⁸ Subsequent FFP transfusions at 10 mL/kg every 12 hours for a minimum of 3 days to maintain FV activity levels greater than 20% has been reported in the literature.⁵⁷

FVII activity does increase during pregnancy, and women with mild FVII deficiency may achieve hemostatic FVII activity levels that are adequate for delivery. However, women with severe FVII deficiency remain at increased risk for hemorrhage and pregnancy loss with persistently low FVII activity. In one report of more than 90 live births in women with FVII deficiency, FVII replacement was administered in only 32% of births. Postpartum hemorrhage occurred in 13% of deliveries with FVII replacement versus 10% of deliveries without FVII replacement.⁵⁸ Given the discrepancy between FVII activity and bleeding phenotype, delivery management of women with FVII deficiency should be decided on a case-by-case basis.⁵⁹ One report suggests consideration of rFVIIa infusion at 15 to 30 μ g/kg every 4 to 6 hours for a minimum of 3 days in women

with a severe bleeding history, FVII activity less than 20% in the third trimester, or before cesarean delivery.⁵⁷

Spiliopoulos and Kadir⁶⁰ carried out a systematic review of obstetric and gynecologic aspects of FX deficiency in 332 patients. A quarter had heavy menses with 66% requiring blood products. Hemoperitoneum was reported in 2.4% with 6 of 8 requiring surgical intervention and 2 of 6 requiring an oophorectomy. Of 31 pregnancies in 19 women with FX deficiency, 30% were preterm births and 38% of the preterm births resulted in neonatal death. Most women who had preterm births were not on FX prophylaxis. Postpartum hemorrhage occurred in 22% of deliveries with one culminating in a hysterectomy. Prophylactic FX infusions via pdFX, PCC, or FFP during pregnancy should be considered if there are systemic bleeding symptoms, vaginal bleeding, retroplacental hematoma, or history of fetal loss. A scheduled or cesarean delivery facilitates coordination of obstetrics, hematology, and hospital pharmacy and should be considered on an individual basis. Factor replacement with pdFX 25 IU/kg, FFP 10 to 15 mL/kg, or PCC 20 to 40 IU/kg should be administered at the onset of labor targeting FX activities of 20% to 40%.⁶¹ Recommendations for management of women with severe FX deficiency through the reproductive years have been presented by Nance and colleagues.⁶²

Research

Given the rarity of RBD worldwide, prospective clinical studies evaluating optimal treatment approaches and novel therapies are challenging to perform and thus remain limited. National and international registries such as the EN-RBD,³⁹ the NARBD registry,²⁸ and United Kingdom Haemophilia Centre Doctors' Organization Registry³⁸ have proved to be critical resources for advancing knowledge on RBD phenotype, genotype, and management. For FII and FV deficiencies, no factor-specific concentrates are currently approved for bleeding management. A solvent/detergent-treated plasma-derived FV concentrate in development has been tested using plasma samples from adult patients with FV deficiency.⁶³ Plasma samples spiked with the FV concentrate demonstrated a linear dose-dependent improvement in standard clotting assays (ie, PT/aPTT), FV activities, FV antigen levels, and thrombin generation profiles. Although this FV concentrate shows great promise, this concentrate is not yet commercially available and further studies evaluating safety and efficacy in individuals with FV deficiency are warranted. Long-acting fusion molecules that seek to extend the short half-life of rFVIIa have been in development and studied preclinically; however, these studies have yet to be investigated or approved in FVII deficiency.²⁹

Gene therapy trials have primarily focused on increasing coagulation protein expression in hemophilia A and B; however, expression of the FVII zymogen using a liver-directed adeno-associated virus serotype 8 has been explored in dogs with FVII activity less than 1%.⁶⁴ FVII-deficient dogs achieved clinically therapeutic FVII activity of 15% or more, which was sustained for more than 1 year without evidence of aberrations in prothrombotic markers. However, further studies are warranted.

SUMMARY

RBD consist of a group of inherited bleeding disorders that share similarities in pathophysiology of disease and clinical presentation but can vary in disease management. Although these disorders affect a small population of individuals worldwide, bleeding symptoms can significantly impact an individual's quality of life and result in adverse bleeding-associated complications. The clinician must remain cognizant that these patients can have the "worst" of both worlds of severe hemophilia and von Willebrand

disease in terms of both deep tissue and mucosal bleeding and major obstetric and menstrual bleeding complications. The clinician must also be aware of risk of bleeding in some heterozygotes. Factor replacement remains the primary treatment modality in each of the RBD. Adjuvant therapies such as antifibrinolytics and hormonal therapies in women with heavy menses greatly supplement the management of bleeding symptoms. Further research addressing the RBD that lack a dedicated, high-purity, single-factor product as well as alternative factor replacement and drug development warrant greater attention. Further clinical care and research regarding indications for prophylaxis in general and in the obstetric and gynecologic setting is also needed.

CLINICS CARE POINTS

- RBD can present with a variety of bleeding symptoms including deep tissue and mucosal bleeding that ranges from mild to severe and can present in the neonatal period into adulthood.
- Evaluation of the PT, aPTT, PT/aPTT mixing studies and factor activity and antigen levels are important laboratory studies for establishing RBD diagnosis.
- Molecular testing of factor genes can aid in the diagnosis but are not mandatory for RBD. Certain mutations may correlate with higher risk of severe bleeds, such as ICH, and may suggest the need for early initiation of factor prophylaxis.
- Plasma factor activity levels do not always correlate with the bleeding risk for some of the RBD. Clinical assessment of bleeding is consequently paramount beyond the level itself with treatment appropriate even in cases of “moderate or mild” factor deficiency.
- Routine follow-up by providers with expertise in managing patients with inherited bleeding disorders or at a facility with a dedicated comprehensive care model of care akin to the care of individuals with hemophilia A and B, in addition to obstetric and gynecologic multidisciplinary engagement, is strongly advised to optimize patient care and outcomes over the age spectrum.

DISCLOSURE

G.B. has received honorarium from Bio Products Laboratory, the manufacturer of plasma-derived factor X concentrate. P.K. has no conflicts of interests to declare.

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